

K. Reshma  
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## SIMATS ENGINEERING ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA1714

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### COURSE OUTCOME COVERED

**CO2:** Experiment search and game-solving techniques such as Greedy Search, A\*, CSP, Minimax, and Alpha-Beta pruning with heuristic functions for solving complex AI problems efficiently. (BL3)

Assessment Tool Weightage: 40%

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QUESTIONS: 3

Duration: 1 Hour  
Total Marks:30

Annexure A

### Dry Run Evaluation

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Code of Conduct:

I K. Reshma (192565062) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.  
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: K. Reshma

20+28+20+18+9  
795

# DRY RUN EVALUATION

K. Rishma

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Artificial

Intelligent

CSA1714

## ① constraint satisfaction problem

### Problem

Four students A, B, C and D must be assigned to 9AM, 10AM and 11AM.

### Constraints

\* A is not at 9AM

\* B is before C

\* D is not at the same time as A

\* C is not at 11AM

### Variables

$$X = \{A, B, C, D\}$$

### Domains

$$D(A) = D(B) = D(C) = D(D) = \{9, 10, 11\}$$



## constraint propagation

since  $c \neq 11$ ,  $c$  can be:

$$C = \{9, 10\}$$

since  $B$  must be before  $c$ :

→ If  $C = 9 \rightarrow B$  would need to be before 9, which is impossible.

→ Therefore  $C = 10$

→ Hence  $B = 9$

Now  $A$  cannot be 9, so:

$$A = \{10, 11\}$$

$D$  must not be the same time as  $A$ .

choose:

$$A = 10 \text{ am}$$

$$B = 9 \text{ am}$$

$$C = 10 \text{ am}$$

$$D = 11 \text{ am}$$

## Valid Assignment

| student | Domain     | constraints | Assigned slot |
|---------|------------|-------------|---------------|
| A       | 9/10/11 AM | $A \neq 9$  | 10 AM         |
| B       | 9/10/11 AM | B before C  | 9 AM          |
| C       | 9/10/11 AM | $C \neq 11$ | 10 AM         |
| D       | 9/10/11 AM | $D \neq A$  | 11 AM         |

### Verification

\*  $A = 10 \neq 9$

\*  $B = 9$  is before  $C = 10$

\*  $D = 11 \neq A = 10$

\*  $C = 10 \neq 11$

$$A = 10, B = 9, C = 10, D = 11$$



②

## Map colouring CSP

### problem

Regions A, B, C and D form a cycle.

### Adjacent constraints:

- \* A is adjacent to B and C.
- \* B is adjacent to A and D.
- \* C is adjacent to A and D.
- \* D is adjacent to B and C.

### Available colours:

$\{R, G, B\}$

### where:

R = Red

G = Green

B = Blue

### Variables

$X = \{A, B, C, D\}$

### Domains:-

$D(A) = D(B) = D(C) = D(D) = \{R, G, B\}$

## Back to coloring Solution

Assign A: Red

Since B and C are adjacent to A, they cannot be Red.

choose:

A: Red

B: Green

C: Green

D: Blue

D is adjacent to both B and C, so D cannot be Green. It can be Red or Blue.

choose

D: Red



## Final Assignment

| Region | Available colours | Assigned colour |
|--------|-------------------|-----------------|
| A      | R/G/B             | Red             |
| B      | R/G/B             | Green           |
| C      | R/G/B             | Green           |
| D      | R/G/B             | Red             |

### Verification

$$A \neq B \rightarrow \text{Red} \neq \text{Green}$$

$$A \neq C \rightarrow \text{Red} \neq \text{Green}$$

$$B \neq D \rightarrow \text{Green} \neq \text{Red}$$

$$C \neq D \rightarrow \text{Green} \neq \text{Red}$$

Therefore:

$$A = R, B = G, C = G, D = R$$

Cryptarithm aka CSP

problem:

$$\text{SEND} + \text{MORE} = \text{MONEY}$$

Each letter represents a unique digit from 0-9.

Also:

$$M \neq 0$$

variables:

$\{S, E, N, D, M, O, R, Y\}$

each variable has the domain:

$\{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$

All letters must have different digits

Step 1: write the addition columnwise

$$\begin{array}{r} \text{SEND} \\ + \text{MORE} \\ \hline \text{MONEY} \end{array}$$



From right to left:

units column

$$D + E = Y + 10C_1$$

Tens column

$$M + R + C_1 = E + 10C_2$$

Hundreds column

$$E + O + C_2 = M + 10C_3$$

Thousands column

$$S + M + C_3 = O + 10C_4$$

Ten-thousands column

$$C_4 = M$$

Since the final result has five digits,  
the carry must be 1.

Therefore,

$$M = 1$$

Step 2: constraint propagation

The standard valid assignment is:

| letter | degree |
|--------|--------|
| S      | 9      |
| E      | 5      |
| N      | 6      |
| D      | 7      |
| M      | 1      |
| O      | 0      |
| R      | 8      |
| C      | 2      |

All degrees are unique:

9, 5, 6, 7, 1, 0, 8, 2

unique degrees

$M \leq 1$ , so  $M \neq 0$



step 3 verify the addition

substitute the digits:

$$SEMD = 9567$$

$$MORE = 1085$$

Therefore:

$$\begin{array}{r} 9567 \\ + 1085 \\ \hline 10652 \end{array}$$

AND:

$$9567 + 1085 = 10652$$

Since:

$$MONEY = 10652$$

the solution is correct.

Find Assignment

$$S=9, E=5, N=6, D=7, M=1, O=0, R=8, Y=2$$

Final Verification

$$9567 + 1085 = 10652$$